Trustworthy Collective Reasoning and Evaluation for AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

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Problem. Large language models (LLMs) increasingly power teams of AI agents: one proposes an answer, others evaluate it, challenge its reasoning, or contribute missing evidence. Yet majority voting and LLM-as-a-Judge scores can be confidently wrong when agents repeat the same error, favor their own outputs, or converge through discussion without independent verification. Consider a naval logistics assistant reporting that a replacement component is available: several evaluator agents endorse its recommendation, but each cites the same stale inventory snapshot while an updated depot record tells a different story. How can a collective distinguish genuine corroboration from repeated mistakes and then correct its reasoning?

Central hypothesis. Inspired by how human teams benefit from complementary expertise and independent dissent yet suffer from conformity and shared misconceptions, we hypothesize that collective AI reasoning becomes more reliable when agents estimate peer expertise, account for dependent judgments, and resolve disagreements through targeted evidence checks rather than additional votes. The goal is not merely a better model ranking: a trustworthy collective should know which claims remain unsupported, improve an answer through critique, and abstain when its apparent consensus is unreliable.

Research plan.

- **Learn reliable peer evaluation.** Build on peer ranking to assess both candidate answers and intermediate reasoning steps. Record independent, blinded judgments before discussion; model task-dependent judge ability, question difficulty, self-preference, and shared-source effects. Infer relative reliability from unlabeled peer comparisons, then calibrate correctness and uncertainty using a small independent labeled set when available. Study when judge weights and dependence are identifiable, and when agreement cannot support an absolute confidence claim.
- **Turn disagreement into evidence-grounded reasoning.** Drawing on human deliberation, investigate when independent dissent among LLM agents reveals reasoning errors rather than amplifies shared mistakes. Develop a structured deliberation method that aligns intermediate claims across independently generated solutions, identifies consequential contested inferences, and selects targeted checks through independent source retrieval, counterexample testing, or tool-based verification. Estimate whether a challenge supplies valid new evidence or merely persuasive, unsupported criticism; use this assessment to decide when agents should revise, retain, or abstain from an answer. Study conditions under which verified minority challenges improve collective accuracy, while limiting harmful revisions of initially correct answers. Test whether expertise-diverse, human-inspired agent profiles yield more independent, verifiable critiques than ordinary agent ensembles, without assuming that simulated diversity reproduces human behavior.
- **Use collective feedback to improve agent decisions.** Learn which peers or checks to consult next based on complementary strengths, uncertainty, and the type of suspected failure. Feed verified corrections and process-level critiques back into agent training so evaluation improves future reasoning, not just a leaderboard. Investigate policies for revision, escalation, and abstention when evidence remains inadequate; computation and interaction cost will be measured, not treated as the research objective.

Evaluation and expected outcomes. Evaluate on MEQA, MuSiQue, and MATH, plus controlled multistep cases with duplicated sources, self-preference, and deliberately correlated errors. Compare with a single strong judge, majority voting, unmodified peer ranking, best-of-N, and unconstrained multi-agent debate. Report agreement with independent reference judgments, ranking stability, confidence calibration, final-answer accuracy after revision, recovery of correct minority answers, harmful answer flips, and evaluation cost. Where human interaction data are available, compare real human disagreements, persona-simulated disagreements, and ordinary AI-agent disagreements to measure critique independence, evidence quality, and downstream corrections. Hold out topics and agent families to test transfer.

Naval relevance. Naval analysts synthesizing maintenance, readiness, and mission-status reports may receive several AI assessments that appear independent but all trace to one outdated record. A collective reasoning assistant could expose the shared source, solicit a genuinely new check, and explain whether the evidence warrants an answer or escalation. This fits Neskovic's interests in large-scale evaluation of distributed information, collaborative problem solving, expertise assessment, and principled decision making under uncertainty.

Prior work and PI qualifications. Our PRD peer-ranking and discussion work (Li et al., 2024, TMLR) provides a foundation for aggregating model judgments; IQA-EVAL (Li et al., 2024, NeurIPS) provides experience with simulated human-AI interaction for evaluation; and FG-PRM (Li et al., 2025, Findings of EMNLP) provides fine-grained feedback on reasoning errors. The PI holds a 2024 NSF CAREER award and received a 2023 Amazon Research Award for process-guided fine-tuning. As a postdoc, the PI led system integration for the RESIN team's RESIN-11 delivery system in DARPA KAIROS.